S5 Glossary

Terms as used in this article, with claim ceilings

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Abstract Definitions for Noetic Diffusion Theory 1.0. Where relevant, entries distinguish the theoretical object, the operational chart estimator, and what is not established.

Reading guide. **Object** = theoretical quantity. **Implemented** = what a current contract actually computes. **Not established** = common over-reading.

A – D

Accessibility (dynamical / noetic). Object: the geometry of the family of protocol-conditional kernels $\{K^h\}_{h>0}$ over future **states at horizon h**. The latent process has a transition law K^X; experiments generally estimate projections or chart-conditional approximations K^x. Unadorned K^h means K^x unless a latent argument is explicit. Path-dependent quantities (committors, routes) are defined under the corresponding process, not by a single endpoint kernel. Not workspace ignition, not report, not a consciousness scalar.

Attractor / basin. Object: long-run organization of the transition law. Implemented: not recovered as unique wells on MNPS here. Visual clustering is not evidence of an attractor.

Committor q. Object: probability of hitting B before A. Implemented: conceptual in NMD; 2-D finite-sample recovery closed in companion simulation.

Control c_t / protocol γ. Object: anything allowed to change b, a, mobility M, potential V, or gates, including a path of c or history H_t. RVC is the minimal special case. Not a unique biological executor.

Diffusion tensor a. Object: local second-moment geometry of stochastic spread, $a = \Sigma\Sigma^\top$. Implemented: truth-known envelopes in restricted linear SDEs; not a default empirical NMD export without residual gates.

E – L

Embodied Anchoring Principle (EAP). Object: bodily context may condition accessibility. Implemented: bounded timing (e.g. SO/respiratory phase); spindle-geometry **efficacy** rejected. Not a third MNPS axis.

FAR (finite-amplitude resilience). Object: whether a finite kick dislodges the state. Implemented: not an empirical NMD export here.

Generator $\mathcal{L}_{t\,|\,c}$. Object: infinitesimal description of the controlled diffusion (main text and S1). Implemented: not fitted to neural data.

HELS / HELS-A / HELS-A1. HELS asks whether history writes V. HELS-A asks whether access can change on frozen V. HELS-A1 is the biological eligibility/estimator gap. Do not collapse the three.

Jacobian J / MNJ. Object: local derivative of drift. MNJ means that Jacobian **on an MNPS chart**. Implemented: operational chart Jacobian. Not an intrinsic manifold tensor; α and ω inherit chart dependence.

M – R

Mobility tensor M. Object: SPD matrix converting a Euclidean gradient into potential-driven drift, $b_{\mathrm{pot}} = -M\nabla V$. If a metric g is declared, $M = g^{-1}$. Not the diffusion tensor a; no fluctuation–dissipation link is currently assumed.

MNPS (3D / stratified 9D). Implemented: versioned observation chart (m, d, e) and a nine-dimensional refinement. Not the latent state, not NDT.

NeuralManifoldDynamics (NMD). Implemented: versioned measurement contract for charts and selected estimators. Not the theory.

Noetic. Scope marker: mind-level state dynamics. Not a claim that coordinates constitute experience.

Potential V. Object: NDT state-dependent landscape in the drift split. Not IIT's Φ.

Reachability (stochastic short-horizon). Object: covariance-based approximation to finite-horizon **predictive spread** under K^h, summarized by a predictive-spread ellipsoid. Implemented in companions as covariance-tube summaries (e.g. effective dimensionality, persistence). Not classical control-theoretic reachability and not Gramian controllability.

Chart-speed coupling. Object: Pearson (or similar) correlation between regional sequences of epoch-to-epoch chart speed. Implemented in one Allen companion as a CA1–VISam omission-minus-expected contrast. Not stochastic short-horizon reachability, not $\Sigma^{(h)}$, and not a Gramian.

RVC (Rhythmic Variance Control). Object: $r(t) \to \sigma(t)$ as minimal accessibility control. Not proven spindle denoising.

S – Z

Spectral / numerical abscissa (α, ω). Chart-level summaries of local linearization (growth of modes; transient amplification). Stable $\alpha < 0$ is not a phenotype if $\omega > 0$.

Thalamic access. Candidate architecture for changing K^x on frozen V. Synthetic operators exist; biology is association, not identification.

Bibliography